Greenbrier Science Olympiad Invitational

Crime Busters B Test

2025-2026

KEY

Team Name:_____

138 points

The Crime:

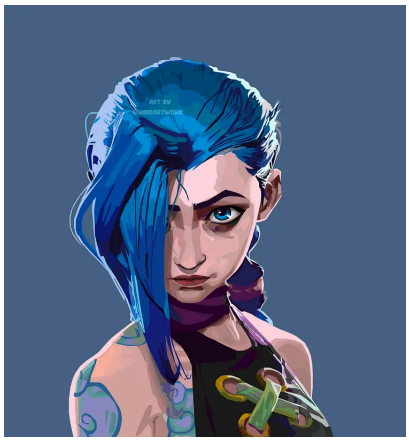
On Wednesday afternoon at Greenbrier Middle School, several students were working in the STEM Makerspace to prepare for the upcoming STEM Expo. The students are competing to win a $50,000 scholarship. Sam was printing his Hand Grip Prototype in the 3D printer, which he made to help individuals with limited hand mobility grasp objects more easily. At the last meeting, Mr. James praised Sam for his design in front of his peers.

At 3:20, Sam's mom came to pick him up early to go to a dentist appointment. Rushing to leave, Sam left his prototype on the front workbench near the 3D printer. Between 3:25 and 4:05, Mr. James briefly stepped out to help another student who was working in the hallway. When he returned to put Sam's Hand Grip up, the prototype was gone.

The area showed some signs of a disturbance.
- A hex wrench on the floor beside the bench
- The rolling chair was repositioned with trace evidence on the wheel.
- Small fragments of plastic
- Visible powder residues on the workbench.
- a small liquid spill near the sink
- light fingerprint smudges on the storage bin and other places.

Five students were confirmed to be in the area or near the area during the period of time Mr. James was not in the room. Each had a legitimate reason to be present and had access to the workstations and tools near the workbench and 3D printer.

 **Vi**	Vi is a hands-on student who often helps assemble or build large projects for the Robotics Club. She frequently wears athletic clothing made out of cotton, and she regularly handles storage bins and heavy tools that have plastic handles. Vi sometimes performs tests that involve powders such as salt and baking soda. Vi also walks through the school courtyard on her way to the Makerspace from her last period class, picking up bits of loam-type soil. Statement to the Police: She was reorganizing tools and helping clean up the space before leaving. Vi has a plain whorl fingerprint.

 **Jinx**	Jinx is a creative but all-over-the-place designer who often experiments with powders, colorful plastics, and often uses the 3D printer. She rides her bike to school and parks her bike in the sandy path along the school building to get to the STEM wing. She often works quickly and leaves materials behind without realizing it. Jinx uses absorbent powders to clean sticky residues from her workstation. Statement to the Police: She was brainstorming a design idea and moving in and out of the room while testing parts. Jinx has an ulnar loop fingerprint.
 **Jayce**	Jayce is really competitive and focused on finishing his project for the expo. He primarily works with metals and various tools. Jayce parks his car in the student parking lot, which is paved. Jayce and Sam were previous project partners until they had a disagreement, which led them to split. He keeps a clean and organized workspace. Statement to the Police: He was checking progress on his part, he was printing on the other 3D machine, and also inspecting his presentation display. Jayce has a plain arch fingerprint.
 **Caitlyn**	Caitlyn is organized and pays attention to detail. She handles small containers and tools on a regular basis. She constantly keeps her areas clean by using rubbing alcohol or hydrogen peroxide. She carries a fabric bag with her project notes and plastic containers with her powders in it. She often walks with Vi to the Markerspace through the courtyard path. Statement to the Police: She was cleaning up Jinx's drink that she spilled on the table where she was working. Caitlyn has a radial loop fingerprint.
 **Ekko**	Ekko is a resourceful builder who moves between the coding station and 3D printer areas. He often grabs miscellaneous tools and plastics for quick tests. He rides his scooter around campus and spends time near the STEM garden (soil is sandier) during lunch, where he plays with the school cat. He explained that he was testing the code of his upcoming project and may have moved some of the tools. Statement to Police: He does remember seeing Sam's prototype on the bench, but did not touch it. He claimed that he and Sam are best friends. Ekko has a tented arch fingerprint.

Qualitative Analysis (72pts)

The forensics team has collected powders and mixtures at the scene. You will need to identify each of the powders, mixtures, and liquids in the table below

P1: was found on the workbench.

P2: was found in the trash can.

P3: was found on a paper towel near the sink.

P4: was found on the tool.

M1: was found in a small shared container used by multiple clubs. (2 powders mixed)

L1: Small spill collected from the sink area.

L2: Sample taken from a paper towel found in the trash.

L3: Residue collected from a handled tool.

1. P1: salt (6pts) Sodium chloride	**2.** P2: baking soda (6pts) Sodium Bicarbonate	**3.** P3: cornstarch (6pts)	**4.** P4: Vitamin C (6pts)
5. M1: sugar & baking soda (7pts)	**6.** L1: lemon juice (6pts)	**7.** L2: hydrogen peroxide (6pts)	**8.** L3: rubbing alcohol (6pts)

9. What is the chemical formula for P1? NaCl (2pts)
10. Who does P1 implicate?: Vi (1pt)
11. What is the chemical formula for P2?: $NaHCO_3$ (2pts)
12. Who does P2 implicate?: Vi (1pt)
13. What is the chemical formula for P3?: $(C_6H_{10}O_5)_n$ or $C_{27}H_{48}O_{20}$ (2pts)
14. Who does P3 implicate?: Jinx (1pt)
15. What is the chemical formula for P4?: $C_6H_8O_6$ (2pts)
16. Who does P4 implicate?:Caitlyn (1pt)
17. What is the chemical formula for L1?: $C_6H_8O_7$ (citric acid) (2pts)
18. Who does L1 implicate?: none (1pt)
19. What if the chemical formula for L2?: H_2O_2 (2pts)
20. Who does L2 implicate?: Caitlyn (1pt)
21. What is the chemical formula for L3?: C_3H_8O (2pts)
22. Who does it implicate?: Caitlyn (1pt)
23. What safety precautions should be taken when testing an unknown powder? (1pts)
    - Wear goggles and PPE
    - Do not taste or smell
    - Properly use the equipment to prevent cross contamination
    - Wash hands

24. Briefly describe the difference between the crystal size of sugar and salt. (1pts)

Sugar crystals are hexagonal and larger; salt crystals are cubic and small

Polymer Testing (15 pts)

A. Plastics

Three plastic fragments were collected from the Makerspace. You have placed your results in the table below. Identify the plastic fragments by their full names and not their initials.

	Flame Behavior	Odor	Residue
P1: found near the prototype stand	Burns with a sooty flame	Sweet smell	Hard with a black bead
P2: found near the tool storage bin	Burns with a light blue flame and drips	Has a waxy and slightly paraffin smell	Soft with a cloudy bead
P3: collected near the sink	Burns with a steady yellow flame	Has a candle-like smell	white, crumbly ash/bead mixture

25. Identify Plastic 1: Polystyrene (2pts)

26. Which suspect(s) does it implicate?: Jinx or Ekko (1pt)

27. Identify Plastic 2: Polypropylene (2pts)

28. Which suspect(s) does it implicate?: Vi and Caitlyn (1pt)

29. Identify Plastic 3: High Density Polyethylene (2pts)

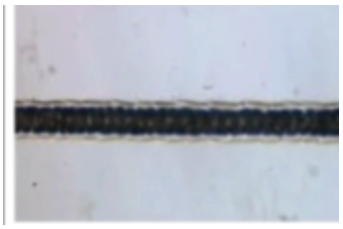
30. Which suspect(s) does it implicate?: Jayce and Caitlyn (1pt)

What color flame does PVC have during a burn test? Green (1pt)

B. Hair Evidence

Identify the following samples of hair that were found on the rolling chair.

 Hair Evidence 1	31. Identify Hair 1 and who does it implicate, if anyone? 1 pt Human None

 Hair Evidence 2	32. Identify Hair 2 and who does it implicate, if anyone? 1pt Animal (cat) Ekko

33. How does the medulla differ between human hair and animal hair? (1pts) The medulla in human hair is thinner and at times fragmented, while the medulla of animal hair is thicker and continuous.

34. Why is hair considered class evidence instead of individual evidence? (1pts) If hair does not have DNA (root) can not be linked to a particular indivudal

C. Fiber Evidence

Identify the fiber that was collected from the scene based on the results of a burn test that was conducted and a picture that was taken of the fibers.

Fiber 1: Melts and burns, and produces a plastic-like smell and a hard bead. Fiber was collected from the handled tool 35. Synthetic Fibers (.5)	
Fiber 2: Burns steadily and smells like burning paper, producing a soft gray ash. Fiber was collected from the workbench. 36. Cotton (Vegetable Fibers) (.5)	

Chromatography Testing (6 pts)

An ink sample was taken from a pen mark found on the workbench near the prototype. You will need to test the ink sample and calculate Rf values. Once completed, you will compare your results to the provided suspect chromatograms. Tape your evidence below and show RF calculations.

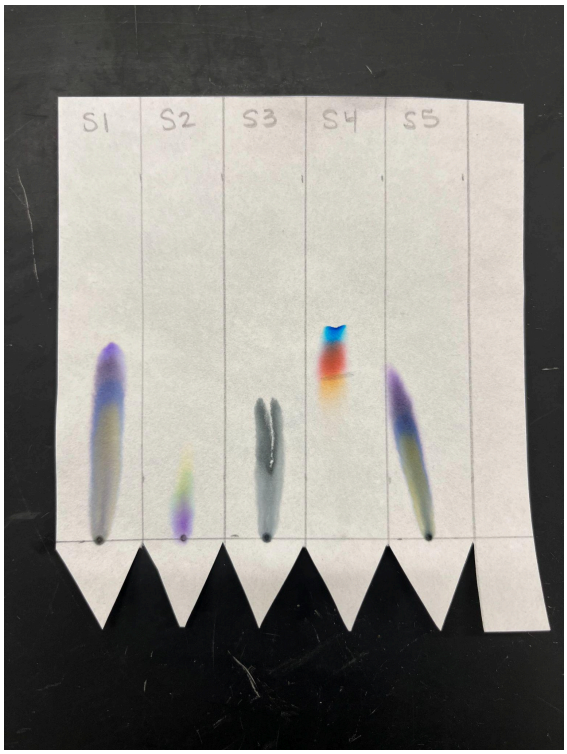
37. Which suspect's pen chromatogram best matches the crime scene sample? Caitlyn (2pts)

38. What does Rf stand for, and why is it important when doing paper chromatography? (1pts)

retention factor; shows the movement of the pigments in the dyes and allows comparison between samples to see similarities or differences.

39. Tape results below (3pts)

Chromatography of suspects was provided below, and the student's results of their chromatography should implicate suspect 4. 1 pt: chromatography results; 2pts: Rf calculations



Physical Evidence (12pts)

A. Fingerprints

The following prints were lifted from the crime scene:

Lift 1: Was lifted from the hex wrench handle found by the prototype stand

Lift 2: Was lifted from the storage bin

Lift 3: Was lifted from the door

Lift 1	Lift 2	Lift 3
		

40. Lift 1 is what type of fingerprint?: Radial loop (1pt)

41. Lift 2 is what type of fingerprint?: Whorl (1pt)

42. Lift 3 is what type of fingerprint?: Ulnar loop (1pt)

43. Lift 1 implicates what suspect?: Caitlyn (.5pt)

44. Lift 2 implicates what suspect?: Vi (.5pt)

45. Lift 3 implicates what suspect?: Jinx (.5pt)

46. If a print is partial, what characteristics should you use to identify it? (1pt) pattern type and minutiae features



Label the minutiae features for the following letters in the diagram

47. A. bifurcation (.5pt)

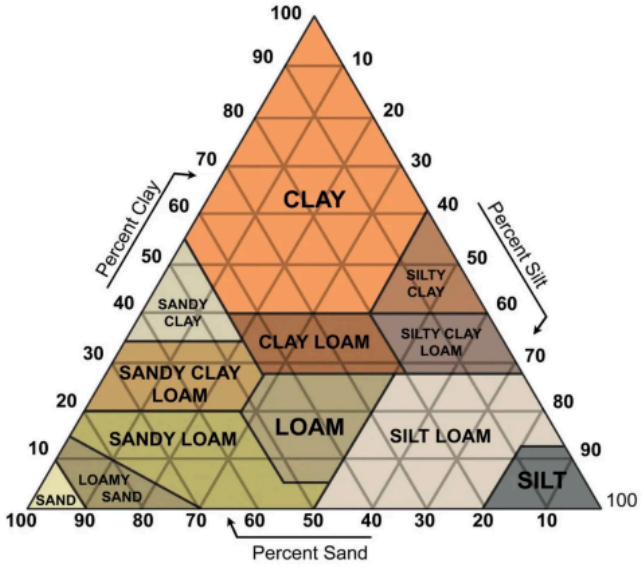
48 B. island (.5pt)

49. C.ridge ending (.5pt)

50. D. Delta (.5pt)

B. Soil Analysis

A soil sample was recovered from the area where the prototype went missing. The result of the soil sample is below.



? Sand	30% Silt	15% Clay

51. What would be the percentage of the sand? 55% (.5pt)

52. What type of soil was found at the crime scene? Sandy Loom (1pt)

53. Which suspect(s) does this type of soil implicate? Catilyn and Vi (1pt)

54. Explain how particle size affects how soil behaves in water? (2pt) Sand, silt, and clay have different particle sizes; therefore, each behaves differently in water. Sand particle sises are the largest and settle the fastest, which allows water to drain quickly. Clay particles are finer and retain water more than sand and silt particles.

Analysis (33pts)

Using all the evidence collected from the Makerspace, determine which suspect most likely removed Sam's prototype.

State one suspect who is the culprit (5pts), and use at least four different evidence categories to help prove that they committed the crime. Explain how each piece supports your conclusion. (20pts)

Also, use evidence to show how the other suspects are not responsible for committing the crime. (8pts)

Caitlyn is the thief:

- Lift 1 fingerprints match Caitlyns, which were found on the hex wrench. The hex wrench was found next to the work bench were the prototype was last seen.
- Chromatography - the ink that was found at the scene matches Caitlyn's pen chromatogram.

- Liquids - rubbing alcohol (found on a tool) and hydrogen peroxide (came from a paper towel in the trash can) are cleaning agents, and she was helping Vi clean areas within the Makerspace. These liquids were found in proximity to where the prototype was taken.
- Soil - She travels through the courtyard, which collects soil on the bottom of her shoes
- Polymers/Plastics: Keeps her powders in plastic bottles and uses the cleaning bottles.

The other suspects did not commit the crime because:

Vi - although she and Catilyn have the same soil sample, the fingerprint lift did not match.
Jinx - soil sample did not match what was found at the scene.
Jacye - although he cleans - none of the other pieces of evidence put him near the scene. Therefore, no matching soil (paved), no matching fingerprint or ink sample.
Ekko - soil found is sanider area, and his fingerprint is a tented arch.

Greenbrier Invitational - Repaired practice edition - Answer key

Division B | Crime Busters | Repair appendix

This appendix contains authored replacements and supplied written alternatives. It does not reconstruct unpublished historical material. For the listed item IDs, use this appendix instead of the archived wording and key.

ANSWER KEY - Review only after submitting.

repair-case. (2 points) Authored written substitute for unavailable source material: An elongated stain’s narrow tail points:

Unit: Fingerprints, DNA, prints, soil, and spatter.
Worked-case data: A scene DNA sample shows peaks at 12 and 15 repeats at location 1, and 8 and 8 at location 2. Suspect X: 12/15 and 8/10. Suspect Y: 12/15 and 8/8. Which suspect matches?
Separate investigation observations:
Scene print: A whorl with two deltas.
Suspect A: Right index finger: loop.
Suspect B: Right index finger: whorl with matching ridge endings.
Suspect C: Right index finger: arch.

A. In a random direction

B. In the direction of travel

C. Back toward the source

D. Straight up

Correct: B